

Wesley Wong

High Schooler in Embedded Autonomous Systems

San Jose, CA | [linkedin.com/in/wesley-wong28](https://www.linkedin.com/in/wesley-wong28) | github.com/Wesley28w | [OpenReview](https://openreview.net/profile/Wesley28w)

EDUCATION

Los Gatos High School

GPA: 4.5 weighted / 4.0 unweighted | Focus in Computer Science, Robotics, ML, and Embedded Systems

Expected May 2029

Los Gatos, CA

EXPERIENCE

Independent Researcher and Solo Author

Robotics, Reinforcement Learning, and Embedded Autonomy

Dec. 2025 – Present

San Jose, CA

- Sole-authored research on reinforcement-learning curricula for robotic manipulation; accepted to the NeurIPS 2026 PhysUnderstand Workshop and endorsed to submit in arXiv cs.LG.
- Developing additional work in multimodal robotics, spiking neural networks, and machine learning, with a journal manuscript under submission.

Bay Area Ambassador

Exea Labs

Aug. 2026 – Present

Bay Area, CA

- Connect aspiring high school researchers with compute resources, publication guidance, research feedback, and STEM support. Plan Bay Area workshops that help students launch research projects.

Co-Founder, Co-President, and Software/AI Director

Los Gatos CubeSat

Jan. 2026 – Present

Bay Area, CA

- Co-founded the school's CubeSat team and led software/AI for the 20-member program; developed Linux and STM32/FreeRTOS flight software.
- Single-handedly secured \$10,000 in initial grant funding and founded school Club.

Director of Software

FRC Team 972 Iron Claw Robotics

May 2025 – May 2026

Bay Area, CA

- Led software for a 50-member FIRST Robotics Competition team; helped earn three awards, including at the state championship, and reach the Houston World Championship Division Finals.

PROJECTS

Functional Safety Project | Python, YOLO, OpenCV, IR Sensing, MyCobot

Sep. 2026 – Present

- Building a Digit-inspired safety stack that combines computer vision and infrared sensing to detect people, estimate risk, trigger shutdowns, and validate fail-safe behavior on a MyCobot 280.

TriBot | Fusion 360, Ansys, STM32, Embedded C

June. 2026 – Present

- Designing a scalable multimodal robot from scratch across mechanical CAD, simulation, circuitry, component selection, and STM32 firmware for ground, aerial, and aquatic operation.

Synopsys Science Fair: GPT-Controlled RL Pipeline | Python, GPT, Isaac Lab

Dec. 2025 – Apr. 2026

- Built an AI-powered engineering pipeline that decomposed robotics objectives into structured tasks and used prompt engineering with multi-stage validation to generate and test RL components; placed third in category at SCVSEFA.

Self-Hosted GPU Cluster | Vast.ai, Google Cloud Run, Redis, vLLM

Aug. 2025 – Dec. 2025

- Built an on-demand AI inference platform with automatic GPU routing, low-latency vLLM serving, Redis caching, and a Cloud Run control layer.

Computer Vision and Generative AI Projects | YOLO, Face ID, ComfyUI, Stable Diffusion

2024 – 2025

- Fine-tuned YOLO models for AUV and identification tasks and built local node-based image/video generation and fine-tuning workflows with ComfyUI.

AWARDS AND HONORS

NeurIPS 2026 PhysUnderstand Workshop Paper Acceptance | arXiv cs.LG Endorsement | SCVSEFA Category Third Place
FIRST World Championship Division Finalist | Three FIRST Robotics Awards | Los Gatos High School Researcher Award

TECHNICAL SKILLS

Languages: Python, Embedded C, Swift, Java

Machine Learning and Robotics: PyTorch, NumPy, Matplotlib, NVIDIA Isaac Lab, reinforcement learning, YOLO, OpenCV, transformers, SNNs, Calculus I/II, Linear Algebra

Embedded and Hardware: Linux, FreeRTOS, Raspberry Pi, STM32, Communication Protocols, Fusion 360, Basic Circuit Analysis, FPV drones

Cloud and Developer Tools: Git, GitHub, CMake, Docker, Google Cloud Run, Redis, vLLM, Vast.ai, LaTeX